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Optional Parameters in Web API Attribute Routing

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Optional Parameters in Web API Attribute Routing

In this article, I am going to discuss **Optional Parameters in Web API Attribute Routing** with some examples. Please read our **previous article** before proceeding to this article as we are going to work with the same example that we started in the **Web API Attribute Routing** article where we discussed the following things.

1. Why we need attribute routing?
2. What is Attribute Routing?
3. How to Implement Attribute Routing?
4. What are the advantages of using Attribute Routing?

Optional Parameters in Web API Attribute Routing and Default Values:

You can make a URI parameter as optional by adding a question mark ("?") to the route parameter. If you make a route parameter as optional then you must specify a default value by using parameter = value for the method parameter.

We are going to work with the same example that we created in our last article. In our last article, we use the following Student Model

```
public class Student
{
    public int Id { get; set; }
    public string Name { get; set; }
}
```

Along with we modify the WebApiConfig class as shown below.

```
public static class WebApiConfig
{
    public static void Register(HttpConfiguration config)
    {
        // Attribute routing.
        config.MapHttpAttributeRoutes();

        // Convention-based routing.
        config.Routes.MapHttpRoute(
            name: "DefaultApi",
            routeTemplate: "api/{controller}/{id}",
            defaults: new { id = RouteParameter.Optional }
        );
    }
}
```

Let's modify the Student Controller as shown below.

```
namespace AttributeRoutingWEBAPI.Controllers
{
    public class StudentsController : ApiController
    {
        static List<Student> students = new List<Student>()
        {
            new Student() { Id = 1, Name = "Pranaya" },
            new Student() { Id = 2, Name = "Priyanka" },
            new Student() { Id = 3, Name = "Anurag" },
            new Student() { Id = 4, Name = "Sambit" }
        };

        // Optional URI Parameter with default value
        // URL: /api/students
        // URL: /api/students/1
        [Route("api/students/{stdid:int?}")]
        public Student GetBooksByID(int stdid = 1)
        {
            return students.FirstOrDefault(s => s.Id == stdid);
        }
    }
}
```

In the above example, **/api/students** and **/api/students/1** return the same resource. Alternatively, you can also specify a default value inside the route template as shown in the below image.

```
// Optional URI Parameter with default value
// URL: /api/students/
// URL: /api/students/1
[Route("api/students/{stdid:int=1}")]
public Student GetBooksByID(int stdid)
{
    return students.FirstOrDefault(s => s.Id == stdid);
}
```

This is almost the same as the previous example, but there is a slight difference in the behavior when the default value is applied.

In the first example (**"{stdid?}"**), here the default value 1 is directly assigned to the action method parameter, so the method parameter will have this value exactly.

In the second example (**"{stdid=1}"**), the default value "1" assigned to the method parameter through the model-binding process. The default model-binder in Web API will convert the value "1" to the numeric value 1.

In most of the cases, unless you have custom model binders in your pipeline, the two forms will be equivalent.

In the next article, I am going to discuss **Route Prefix in Web API** Attribute Routing. Here, in this article, I try to explain the Optional Parameters in Web API Attribute Routing step by step with some examples. I hope this Optional parameter **in the Web API Attribute Routing** article will help you with your need. I would like to have your feedback. Please post your feedback, question, or comments about this article.

Dot Net Tutorials
About the Author: Pranaya Rout

Pranaya Rout has published more than 3,000 articles in his 11-year career. Pranaya Rout has very good experience with Microsoft Technologies, Including C#, VB, ASP.NET MVC, ASP.NET Web API, EF, EF Core, ADO.NET, LINQ, SQL Server, MYSQL, Oracle, ASP.NET Core, Cloud Computing, Microservices, Design Patterns and still learning new technologies.

2 thoughts on "Optional Parameters in Web API Attribute Routing"

GISER
DECEMBER 15, 2021 AT 6:44 PM

that's simple and easy to understand, but i have a confused point that in the above example there is a question mark ("?") ,however the other example doesn't have ("?"). So,my question is the code without the ("?") is still the Optional Parameters? My english is not so well,thanks your reply.

Reply

GISER
DECEMBER 15, 2021 AT 7:17 PM

The above example without http verb, how the webapi match the action with "get" or" post" request .
My mean is that if an action want to be called by the get method rather than post method, is there any necessity to add "[HttpGet]"above the method?

Reply

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